Report R17 — The Genome-Wide Null Was Not a Supervision Artefact

R15 concluded that its genome-wide negative came from supervising the wrong target. Re-running the identical scan on a methylation-supervised representation does not reverse it. This report corrects R15.

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Summary

R15 asked whether H&E morphology carries methylation information. Its first arm supervised an attention-MIL on subtype, then asked the resulting embedding about methylation, and found nothing: genome-wide median $\rho = 0.006$, 110,212 of 399,579 CpGs above a permutation null, against 220,925 for the subtype label by itself. Supervising the same architecture directly on six aggregate methylation targets recovered partial ρ 0.135–0.244, and R15's audit concluded that the first negative had been *an artefact of our own supervision choice*.

R15 was explicit that it had not tested this genome-wide — its limitations state that "per-CpG prediction under methylation supervision was not attempted". This report attempts it.

The conclusion does not survive. On the identical scan — same 760 patients in the same order, same site-grouped folds, same ridge, same probe filter, only the embedding changing — the methylation-supervised representation puts **83,369** CpGs above the null (20.9%) against **90,137** for the subtype-supervised one (22.6%). It is not better. It is slightly worse, and its median ρ of 0.0073 against 0.0057 is the same flat nothing.

Verdict: ARM1_NEGATIVE_ROBUST_TO_SUPERVISION, with the secondary verdict **GLOBAL_AXIS_ONLY**. Both were pre-declared as mechanical rules in the config, hashed at `4779152747f14818...` before any arm was computed.

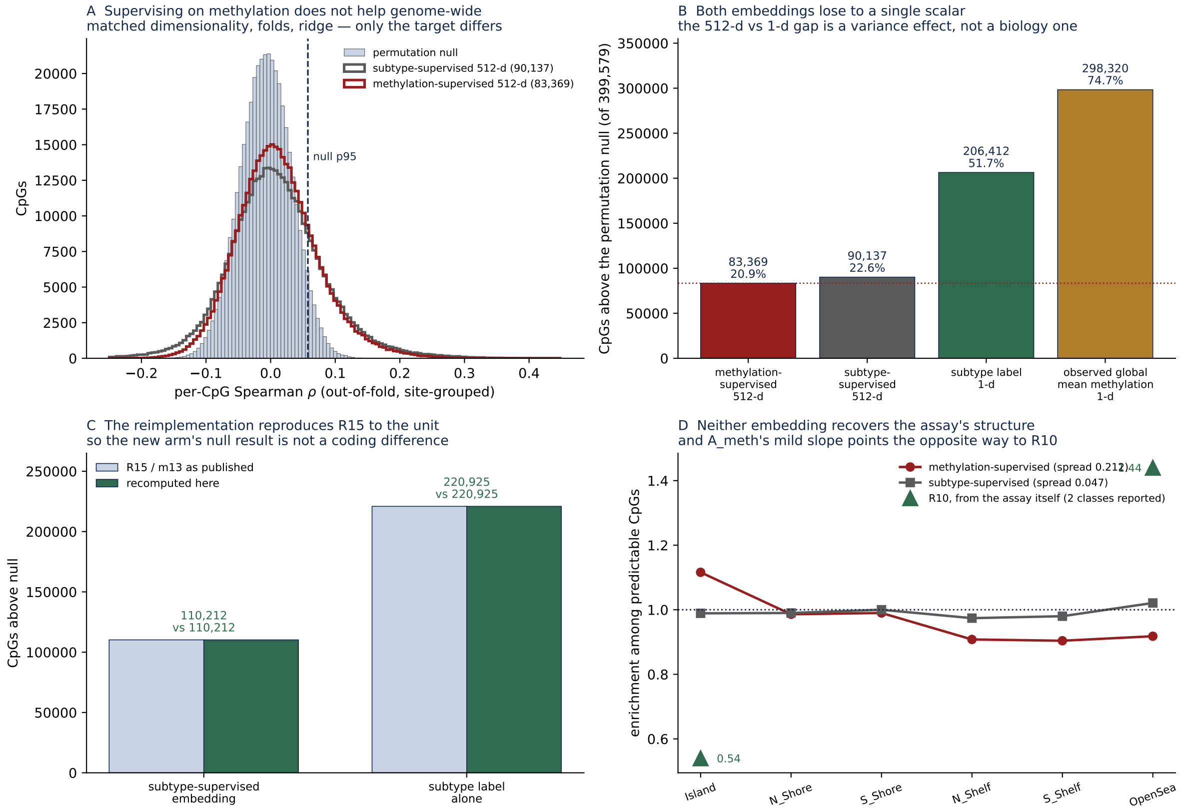
So R15's aggregate-target result stands and its *explanation* of the negative does not. What the methylation-supervised model learned was the aggregate targets, not the methylome.

1 Why the comparison is trustworthy before reading it

The whole claim rests on a reimplementation, so the reimplementation was gated against the original before the new arm was allowed to count.

- **The vectorised Spearman is exact.** R15 looped `scipy.stats.spearmanr` over 399,579 CpGs; this run rank-transforms and takes Pearson, which is equivalent. Gate G5 checks it against scipy on 200 real columns: maximum absolute deviation 2.1×10^{-8}.
- **It reproduces R15's per-CpG vectors.** Gate G4 requires $r > 0.999$ against R15's saved `per_cpg_rho` arrays. Observed: $r = 1.000000$ for both the subtype-supervised embedding and the subtype label.
- **It reproduces R15's headline counts to the unit.** On R15's own null threshold (0.04605): 110,212 and 220,925, exactly the published numbers (Figure C).
- **The folds nest.** Rather than re-deriving cross-validation folds, this run takes `fold_of` as written by the MIL itself, and gate G1 verifies it equals `GroupKFold(5)` grouped by tissue-source site and is identical across both arms. The ridge folds are therefore the same folds the representation was trained under, so a patient's embedding never comes from a model that saw that patient.

Only after all four passed was the new arm computed.



A The decisive comparison: matched dimensionality, folds and ridge, only the supervision target differing. **B** All four arms. **C** The reimplementation against R15 as published. **D** Genomic context of the predictable CpGs.

2 Results

Arm	Predictor	dim	above null	%	median ρ
A_meth	methylation-supervised embedding	512	83,369	20.9	0.0073
A_sub	subtype-supervised embedding	512	90,137	22.6	0.0057
A_sublabel	LUAD/LUSC label	1	206,412	51.7	0.0631
A_globalmean	*observed* global mean methylation	1	298,320	74.7	0.2087

Threshold: 95th percentile of the permutation null, 0.05763; null median −0.0071.

2.1 The two 512-d arms are the finding

They differ in one respect and agree in outcome. Supervising on methylation moves the count from 90,137 to 83,369 — in the wrong direction — and the median ρ from 0.0057 to 0.0073, which is a difference between two numbers that both mean “no relationship”. The subtype-supervised arm even attains the higher maximum, 0.485 against 0.350.

2.2 The 1-d arms are not a fair comparison, and say something anyway

Both scalars beat both embeddings by a wide margin. That gap should *not* be read as biology: a ridge with 512 correlated dimensions fitted on roughly 608 training patients is a higher-variance per-CpG predictor than one well-chosen scalar, so some of the gap is regularisation. R15 saw the same pattern and read it as evidence that the representation was wrong for the question; the honest reading is that it is partly an artefact of dimensionality, which is why this report rests on the matched-dimension comparison instead.

What the scalars do establish is a ceiling. **Knowing a patient's true global mean methylation puts 74.7% of CpGs above null on its own.** Most of what "per-CpG predictability" measures in any arm is one global axis, not CpG-resolution structure. This is the `GLOBAL_AXIS_ONLY` verdict, and it is the reason the six aggregate targets in R15 could improve while the methylome as a whole did not: five of the six are means over large annotation classes, and the sixth is the global mean itself. A representation can learn that scalar and still carry nothing at CpG resolution — which is what the two 512-d arms show it does.

2.3 Genomic context does not recover the assay's structure

R10, working from the array directly, found the KEAP1 methylation phenotype strongly structured by genomic context: islands depleted at 0.54, open sea enriched at 1.44, a spread of 0.90. R15's subtype-supervised arm was flat, 0.97–1.01, spread 0.047.

The methylation-supervised arm is mildly structured — spread 0.212 — but *in the opposite direction*: islands *enriched* at 1.116 and open sea *depleted* at 0.918. Whatever this is, it is not a weakened version of R10's pattern, and it should not be presented as partial recovery.

3 What this changes in R15

R15's measured numbers are untouched. Two of its written statements are not.

- "Training the same architecture directly on methylation *reversed the answer*" and "supervising on methylation directly *recovered the signal*" — true of the six aggregate targets, false genome-wide. The reversal is specific to the aggregates.
- "That flatness of genomic context . . . was in fact evidence that the representation contained no methylation-relevant structure to find" — this does not hold, because the representation that *does* contain methylation-relevant structure produces a genomic context that is still nearly flat, and tilted the wrong way.

R15's audit generalised from six aggregate summaries to the methylome. Its own limitations section named the gap correctly and the gate that would have caught the over-reach was simply never run. That is the transferable lesson: the limitation was *stated*, and stating a limitation is not the same as bounding a claim by it.

4 Limitations

This does not show H&E carries no methylation information. R15's aggregate result — partial ρ 0.135–0.244 beyond subtype, surviving purity adjustment — stands. The claim bounded here is the genome-wide, per-CpG one.

One architecture, one ridge, one alpha. Both 512-d arms use the same gated attention-MIL and $\alpha = 100$. A different readout, or per-CpG hyperparameter selection, is not excluded — though it would have to be pre-declared to mean anything.

The circularity runs the other way here. The methylation-supervised embedding was trained on targets that include the mean over every scanned probe, which should if anything *favour* it in this scan. It still did not win. That makes the negative harder to explain away, not easier.

The null threshold changed slightly. R15 drew two independent permutations, scoring $Y[p_1]$ against a model fitted on $Y[p_2]$; still a valid null, but not the single-permutation null it reads as. Both are reported (p95 0.05763 single, 0.05953 two-permutation) and the choice does not affect the ordering of the arms.

Site-grouped folds only. No random-fold variant was run, deliberately: the leakage question was settled in R15 and re-opening it here would add a comparison nobody needs.

Reproducing this report

`evidence/m14_percpg_meth_supervised.py` (the whole run), `evidence/config.yaml` (pre-declared rules, SHA-256 `4779152747f14818...`), `evidence/gates.json` (13 gates), `evidence/results.json`, `evidence/log.jsonl`, six `per_cpg_rho_*.npy` vectors (399,579 each), `evidence/probes_scanned.txt`. Figure: `python3 make_figure.py`.